Insights Memo: E-commerce Funnel Analysis

Team Beta, Week 7 | SIDONPAY Data Analyst/Scientist Internship | Prepared by: Ahmed Abdullahi Olamilekan

1. Methodology

Data: 13,125 GA4-style events from 3,774 users (raw_events.csv). All aggregation was done in SQL because the full data is too large for Excel and the important (needed) data was only imported to the SQL.

- **Unique events.** One row per user_pseudo_id per event_name, keeping the earliest event_timestamp (ROW_NUMBER). Timestamps are rounded to ~100 seconds, so ties are broken deterministically. Result: 8,530 rows, 12 columns.
- **Top 3 countries.** Ranked by total event count: United States (5,859), India (1,378), Canada (864).
- **Funnel.** Six stages: page_view, view_item, add_to_cart, begin_checkout, add_payment_info, purchase. Each stage is the number of distinct users who triggered it. page_view is step 1 because it is the widest top-of-funnel event (2,075 users).
- **Metrics.** perc_drop is the % of step 1 remaining (as the task defines it); step-to-step conversion is added for diagnosis. SQL results were cross-checked against pandas and match exactly.

2. Key findings

Stage	Full	US	India	Canada	Full % of step 1
page_view	2,075	915	201	158	100%
view_item	392	176	35	28	18.9%
add_to_cart	90	38	6	8	4.3%
begin_checkout	40	17	5	6	1.9%
add_payment_info	23	5	6	4	1.1%
purchase	4	3	0	1	0.19%

- **Biggest leaks are at the top.** 81% of visitors never view a product, and 77% of product viewers never add to cart. Together these account for most of the loss.
- **Checkout is relatively healthy.** 44% of cart users begin checkout and 57% of those add payment info.
- **The last step is a cliff.** Only 4 of 23 payment-info users purchase (17%).
- **Country differences.** The US supplies 44% of page_view users but 75% of purchases; Canada has the best end-to-end rate (0.63%, 1 of 158); India has 0 purchases despite 6 payment-info users.
- **Traffic medium does not explain the top drop.** page_view to view_item is 19.5% for none, 18.1% for cpc and 18.5% for other, so paid traffic is not clearly lower-intent.

3. Insights and hypotheses

These are hypotheses to test, not proven causes.

- **H1: Landing pages do not lead shoppers to products (page_view to view_item).** Only 392 of 2,075 page_view users ever view a product, which may point to homepage browsing with weak product discovery. Test: compare landing-page types and the share of sessions with view_promotion/select_item before view_item; improve category navigation and search (view_search_results is used by only 74 users).
- **H2: Price or shipping uncertainty stops add_to_cart.** 77% of product viewers leave before the cart, and the cart-to-checkout step is comparatively strong, so those who commit tend to proceed. Test: show shipping cost and delivery time on the product page, and A/B test price anchoring and reviews.

- **H3: Payment failure or missing local payment options cause the final-step loss, especially in India.** 6 India users reached payment info and none purchased, and overall only 17% convert there. Test: log payment errors, check gateway support for local methods/currencies, and review India transactions manually before drawing a conclusion.

4. Limitations and next steps

Purchase count is only 4, so country rates are indicative and not statistically reliable. Counts are independent distinct users, not strict sequences: 5 of 23 add_payment_info users never logged begin_checkout, so some step ratios can exceed 100% (India: 6 vs 5). Next: extend to a longer date range, build sequential (session-level) funnels, and slice by device, landing page and campaign (see the Other slices sheet in the workbook).